mPractical No 3

Title: Implementation of PageRank Algorithm for Ranking Web Pages

Aim:

The objective of this project is to implement the **PageRank algorithm**, a web page ranking algorithm that assigns a numerical importance score to each page based on its hyperlink structure. This algorithm is used by search engines to rank web pages according to their relevance and importance in relation to user search queries.

Prerequisite:

1. Knowledge and Skills:

- **Graph Theory:** Understanding of directed graphs, nodes, and edges as the PageRank algorithm models web pages and links as a directed graph.
- **Linear Algebra:** Familiarity with matrix representation of graphs and eigenvectors, as the PageRank algorithm involves matrix computations.
- **Probability and Markov Chains:** Basic understanding of probability distributions and random walks is useful, as the algorithm can be interpreted as a random walk over the web graph.
- **Programming Skills:** Proficiency in Python for implementing the algorithm, along with knowledge of relevant libraries such as NumPy and networkx for graph representation.

2. Tools and Libraries:

- **Python:** Programming language for implementing the algorithm.
- **NumPy:** For efficient matrix operations.

.

Input:

The input for the PageRank algorithm is a directed graph representing a collection of web pages and the hyperlinks between them. The graph is constructed where:

- **Nodes:** Represent web pages.
- **Edges:** Represent hyperlinks from one web page to another.

Example Input:

A simple web graph with 4 web pages and links:

- **Page A** links to **Page B** and **Page C**.
- **Page B** links to **Page C**.
- **Page C** links to **Page A**.
- **Page D** links to **Page C** and **Page B**.

This graph can be represented as an adjacency matrix or adjacency list.

Adjacency Matrix Representation:

```
   A  B  C  D
A [ 0, 1, 1, 0 ]
B [ 0, 0, 1, 0 ]
C [ 1, 0, 0, 0 ]
D [ 0, 1, 1, 0 ]
```

Outcome:

The output of the PageRank algorithm is a ranked list of web pages, where each page is assigned a numerical score based on its importance. Pages with higher scores are considered more "important"

Expected Outcome:

Given the input graph, the algorithm will output the PageRank score for each page:

- **Page A:** 0.35
- **Page B:** 0.25
- **Page C:** 0.30
- **Page D:** 0.10

Theory:

1. Overview of the PageRank Algorithm:

PageRank is a link analysis algorithm developed by Larry Page and Sergey Brin in 1996 to rank web pages based on their importance. The core idea is that a page is considered important if it is linked to by many other important pages.

2. Key Concepts:

a. Graph Representation:

- **Directed Graph:** The web is represented as a directed graph where web pages are nodes, and hyperlinks between pages are directed edges.
- **Adjacency Matrix:** A matrix where each element represents whether there is a link from one page to another.

b. Rank Propagation:

- Each page distributes its rank equally among the pages it links to. The rank of a page is determined by the ranks of the pages linking to it.

c. Damping Factor:

- The algorithm introduces a **damping factor (d)**, typically set to 0.85, which represents the probability that a random surfer continues clicking on links. The remaining 1 - d probability is used to account for random jumps to any page.

3. PageRank Calculation:

The PageRank of a page is given by the equation:

$PR(pi)=1-d / N + d \sum pj \in M(pi) PR(pj) / L(pj)$

Where:

- PR(pi) is the PageRank of page i,
- N is the total number of pages,
- D is the damping factor (typically 0.85),
- M(pi)is the set of pages linking to pi
- L(pj) is the number of outbound links from page pj.

4. Iterative Computation:

- The PageRank algorithm iteratively updates the ranks of pages until convergence. Starting with an initial rank (usually 1/N for all pages), the algorithm recalculates ranks using the above formula until the difference between successive iterations is below a threshold.

5. Convergence:

- PageRank is guaranteed to converge under certain conditions (e.g., using the damping factor), meaning that after enough iterations, the ranks of pages stabilize.

Conclusion:

The implementation of the PageRank algorithm demonstrates how a graph-based approach can be used to rank web pages based on their importance. By simulating a random surfer's behavior across the web graph and iteratively adjusting the ranks, the algorithm provides a numerical value for the relevance of each web page. This project offers insights into link analysis and its real-world application in search engines.